

Temporal-Support Identifiability in Event-Camera Detection Evaluation Pipelines

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Abstract

001 *An event-camera detection benchmark scores each sample*
 002 *at one nominal timestamp, while a recurrent detector forms*
 003 *its prediction from a history whose length the benchmark*
 004 *never records. We formalize the missing requirement as*
 005 *temporal-support identifiability: the variables a protocol*
 006 *records with a score must determine the temporal evidence*
 007 *behind every scored prediction. Because this is a unique-*
 008 *ness property, one controlled pair of runs with identical*
 009 *records and different supports refutes it, and we construct*
 010 *that pair on the released SSM-ViT evaluation. With frames,*
 011 *weights, labels, evaluator and evaluated timestamps fixed,*
 012 *moving only the chunk boundary of the streaming evalua-*
 013 *tion changes the recurrent history behind the same 3 574*
 014 *frames from 1–4 to 17–20 windows and changes the base*
 015 *checkpoint (S5-B) by 4.20 ± 0.36 mAP over all labeled*
 016 *boxes and by 5.23 ± 0.49 mAP on the velocity-evaluable*
 017 *subset, with a graded response at intermediate shifts and an*
 018 *exact zero at zero shift. The same intervention leaves RVT*
 019 *within 0.05 mAP while its state is carried and moves it by*
 020 *$5.39\text{--}6.91$ mAP once the state is reset, so the effect follows*
 021 *the reset policy and not the architecture. The constructive*
 022 *counterpart is Support-Conditioned AP, which scores every*
 023 *checkpoint from the same reset state at a declared history of*
 024 *H windows. On five released Gen1 checkpoints scored on*
 025 *the same 3 000 frames, S5-B is fourth in the pooled score*
 026 *and first at every $H \in \{1, 5, 10, 21\}$, RVT- b overtakes it*
 027 *between $H = 21$ and $H = 40$, three pairwise gaps reverse*
 028 *sign between the pooled score and $H = 1$ with both in-*
 029 *tervals excluding zero, and Kendall’s τ against the pooled*
 030 *ordering is at most 0.00 for $H \leq 21$. Audits on Gen1 and*
 031 *DSEC show that input timing and label-acquisition timing*
 032 *are left implicit by the same timestamp. Five fields recorded*
 033 *with the labels make the support behind a score identifiable.*
 034

1. Introduction

Two recurrent detectors evaluated at the same nominal timestamp are not comparable in time unless the history each one actually used is also identified.

An event-camera detection benchmark defines a sample by a nominal timestamp. A prediction $\hat{s}$ is scored against the ground truth s^* carried at query time t through a function $m(\hat{s}, s^*)$ of position, shape and class. Beyond selecting the label at t , that function contains no variable for the temporal evidence behind either argument, so any temporal discrepancy enters the score disguised as displacement.

A recurrent detector does not read only the instant the benchmark names. It computes $\hat{s}$ from an input window and from state carried across earlier windows. The label it is compared against comes from an exposure of finite width, an annotator or tracker working on the exposed image, and a geometric transfer into another sensor’s frame. The evidence on both sides is an interval, while the protocol records an instant.

We use four terms. The *nominal timestamp* is the instant at which the protocol pairs a prediction with a label. The *effective temporal support* is the past that can influence the prediction, including state carried by recurrence. The *label-acquisition support* is the interval over which the label’s evidence was acquired. The *protocol record* is the set of variables a benchmark specifies and a paper reports with a score: frames, labels, weights, evaluator, input window, nominal timestamps and published acquisition metadata. A protocol is *temporal-support identifiable* when its record determines the effective temporal support of every scored prediction uniquely (Sec. 3). A quantity that can be reconstructed from released code or index files, but is absent from the record, does not meet this condition: two evaluations that agree on the entire record can still differ in it, and nothing reported with the score tells them apart.

Prior work on timing in detection takes the benchmark’s temporal index as given and changes what is measured against it: latency is folded into the score [7, 27, 32], delay is scored separately [8], and temporal windows are re-

designed [21, 23, 25]. None of these asks whether the index determines the evidence behind the number reported under it. That question has a sharp form. Identifiability is a uniqueness property, so it is refuted by a single pair of runs that agree on the whole record and differ in support, and it is restored by recording the variables that fix the support.

We build the refuting pair on the released SSM-ViT evaluation pipeline, using its base and small checkpoints (S5-B, S5-S). In that implementation the nominally carried state does not enter the outputs across evaluation chunks. The recurrent history available to a prediction is a deterministic function of chunk position, and ranges from 1 to 21 windows inside one pooled score. Moving every chunk boundary 5 windows later re-scores the same frames under a longer history at the same evaluated timestamp, with frames, weights, labels and evaluator fixed (3 574 labeled frames scored twice, cluster bootstrap over 406 sequences; $\pm$ denotes one cluster-bootstrap standard error).

Effective temporal support alone changes S5-B by 4.20 ± 0.36 mAP over all labeled boxes and 5.23 ± 0.49 mAP on the velocity-evaluable subset (Sec. 4). The second checkpoint of the same released pipeline, S5-S, moves by 4.10 and 4.81 points. The response is graded in the amount of history returned and is exactly zero at zero shift. The same intervention on RVT leaves it within 0.05 points while its state is carried and moves it by 5.39 and 6.91 points once the state is reset at the boundary. The effect therefore belongs to the reset policy, a protocol variable, and not to one implementation: wherever a protocol does not record the support behind a score, the reported score is underdetermined. For scale, the frequency-generalization result that motivates the SSM-ViT design is a 3.31 mAP drop [23].

The pair has a constructive counterpart. Support-Conditioned AP scores a checkpoint from a reset state with the recurrent history fixed to a declared number of windows H (Sec. 6). It needs no retraining and no new labels, only the released checkpoints. Applied to five released Gen1 checkpoints on 3 000 common frames, it orders them differently from the pooled score: S5-B is fourth in the pooled score and first at each of $H = 1, 5, 10, 21$, and for three pairs the pooled and $H = 1$ gaps have intervals that exclude zero on opposite sides. RVT-b minus S5-B is $+3.13$ points in the pooled score, -4.15 at $H = 1$ and $+4.79$ at $H = 80$, and the two checkpoints exchange rank between $H = 21$ and $H = 40$.

All measurements use released checkpoints, released code and public data: five checkpoints of two architecture families on the Gen1 validation split (20 296 labeled frames, 406 sequences) and eighteen DSEC sequences (21 142 frames).

Contributions.

- *Temporal-support identifiability*, an evaluation property

with two operational criteria: a witness pair that refutes it and a record that restores it (Sec. 3).

- A same-frame intervention that isolates recurrent support. Only the chunk boundary changes, and the S5-B score moves by 4.20 mAP overall and 5.23 mAP on the velocity-evaluable subset, with a dose response, a zero-shift control, a permutation null, and an RVT control measured in both state regimes (Sec. 4).
- *Support-Conditioned AP*, a protocol that fixes the reset state and the history behind each score. On five released checkpoints the leader depends on H (S5-B up to $H = 21$, RVT-b from $H = 40$) and three pairwise gaps reverse sign with intervals that exclude zero (Sec. 6).
- A five-field record that a benchmark can ship with its labels, filled in for two released evaluations, and predictor-side and label-side audits on Gen1 and DSEC of what the timestamp leaves implicit (Secs. 5 and 6). The evaluation scripts, the boundary-shift driver and the per-frame access logs behind every table will be released.

2. Related work

Sensor and runtime latency. Work on event-camera timing takes the benchmark’s temporal index as given and improves what is measured against it. Sensor latency is calibrated away [22, 26], compute latency is folded into average precision [7, 27, 32], delay is scored separately [8], and temporal tolerance has been proposed [33]. Each changes what the predictor does at a fixed benchmark index, or how late its answer may arrive. The object measured here is that index itself and the support a score reported under it pools: we test whether a released benchmark specification uniquely determines the temporal quantities behind its score. Streaming metrics move the instant at which a prediction is matched. Support-Conditioned AP holds that instant fixed and controls how much past the prediction was allowed to use, a variable those metrics leave free.

Temporal windows and event benchmarks. RVT and SSM-ViT redesign or parameterize temporal windows [21, 23], and adaptive-window methods compare alternative partitions [25]. Gen1, 1 Mpx, DSEC-Det and Ev-3DOD use asynchronous measurements, rectified frames, or interpolated labels [14, 16–18], and the event count inside one labeled box can be too small to support detection [2]. Event-
RGB frequency mismatch and exposure duration have also been noted [19, 20, 34]. LEOD [24] states that event detectors take only events triggered before the query time and builds a time-flip augmentation on that premise. It does not record where inside that window the evidence sits. Both released detector implementations leave a frame’s position inside the evaluation chunk unspecified by the protocol. The recurrent state entering a chunk is the state saved at the end of the previous chunk, and is reset only at the

first sample of a sequence. Under streaming sampling every timestep emits predictions, while otherwise only the last does (RVT modules/detection.py:232, 243 and SSM-ViT :315, 318). These works vary the window a detector reads. None reports the recurrent history behind a score, and none compares detectors at a matched history.

Benchmark reliability and evaluation analysis. Prior work studies detector calibration, stratified model evaluation, label-induced benchmark instability, and event-vision evaluation protocols [28–31]. These analyses examine uncertainty or error in reported performance, whereas our focus is the temporal support and acquisition semantics that a nominal detection timestamp leaves implicit.

Label time and attribution. Waymo and annotation studies estimate per-object capture or annotation offsets [3–5]. Event datasets expose per-box time fields, but their degrees of freedom are not necessarily specified. LET-3D-AP and TIDE/DETAD/HOTA decompose or bucket spatial error while taking the label clock as given [1, 9–12].

3. Temporal-support identifiability

Setup. An evaluation run e is everything executed to produce a score: the weights, the data, the evaluator, and the driver that feeds windows to the detector and manages its recurrent state. The protocol record $R(e)$ is the part of e that the benchmark specifies and a paper reports. Let i index the scored predictions and r_i the window at which prediction i is emitted. The effective temporal support $S_i(e) \subseteq \{0, \dots, r_i\}$ is the set of input windows whose content can change prediction i with the rest of e held fixed, and $m(e)$ is the reported score.

Definition. A protocol is *temporal-support identifiable* if, for any two runs e, e' it admits,

$$R(e) = R(e') \implies S_i(e) = S_i(e') \quad \text{for every scored } i. \quad (1)$$

Criterion 1: a witness pair refutes it. Eq. 1 quantifies over all admitted runs, so one pair with $R(e) = R(e')$ and $S_i(e) \neq S_i(e')$ for some i shows that the protocol is not identifiable. If in addition $m(e) \neq m(e')$, the score is not a function of the record, and a reader holding R can neither reproduce m nor say what temporal evidence it summarizes. The refutation is not a population claim. The number of frames and sequences enters only through the precision with which $m(e) - m(e')$ is estimated. Sec. 4 constructs such a pair and measures that difference.

Criterion 2: a record that restores it. Consider a causal detector driven window by window whose recurrent state is cleared at reset points, and let $b_i \leq r_i$ be the last reset at or before r_i . Nothing before b_i can reach prediction i , so $S_i \subseteq \{b_i, \dots, r_i\}$, an interval of $h_i = r_i - b_i + 1$ windows. The support spans that whole interval when the oldest admitted window reaches the output, which a replacement probe on that window checks. A record that adds the reset policy, which fixes b_i , and the history length h_i therefore determines S_i . Two points follow for practice. First, b_i has to be measured on the running evaluation and not read from a configuration: the SSM-ViT release declares a carried state that never reaches its outputs (Sec. 4), so its realized resets are not its declared ones. Second, fixing $h_i = H$ for every prediction and every detector yields scores whose support is identical by construction. This is Support-Conditioned AP (Sec. 6).

4. Primary evidence: the same-frame chunk-boundary intervention

The state the SSM-ViT release carries between evaluation chunks does not affect any of its outputs. It enters as a multiplicative factor on the first coefficient of an associative scan on which the outputs do not depend. Under the released streaming evaluation this bounds the recurrent history available to a detection to the current 21-window chunk. That history ranges from 1 window to 21 with the detection’s position in the chunk, and the reported number averages over all 21 positions. Chunk position is a deterministic function of the release’s own index files and is recorded nowhere in the evaluation protocol.

The mechanism, measured. The property is read from the source and then checked on the running model. Zeroing the state that enters a chunk changes the emitted detection tensor of SSM-ViT by a relative L_2 of 0.00000 at every position of a 8-window chunk. The same test on RVT- τ gives 0.0616 at the first position and 0.0304 at the last. A model that ignored its history would return zero in both cases, so the control ablates the previous window inside the same chunk: SSM-ViT then changes by 0.0276. It is recurrent within a chunk and loses its state only across chunks (Supplement Sec. 2). Repeated with a 21-window chunk, the test changes the output of all three RVT checkpoints and of a third released detector (EvRT-DETR) and changes that of neither SSM-ViT checkpoint (Supplement Sec. 2).

The same-frame intervention. The intervention moves every chunk boundary 5 windows later and changes nothing else: the same frames, the same weights, the same labels, the same evaluator, and the evaluated timestamp of each frame unchanged. 3 574 labeled frames are scored twice

under this shift, once with 1–4 windows of recurrent history and once with 17–20. They are compared paired within frame with a cluster bootstrap over the 406 sequences. The long-history arm gains 5.23 ± 0.49 mAP points for S5-B ($z = 10.7$) and 4.81 ± 0.52 for S5-S ($z = 9.3$) on the velocity-evaluable subset, and over every labeled box the gains are 4.20 ± 0.36 and 4.10 ± 0.32 points. Mean detection confidence, which uses no labels, moves with it by 0.0579 ± 0.0048 and 0.0509 ± 0.0035 . Fig. 1 (bottom) draws one such pair. No quantity the protocol records distinguishes the two arms, so the pair is the witness of Criterion 1.

Graded response and what moves. The size of the boundary shift is a free parameter. Re-scoring the same 3 574 frames after shifting the boundary so that the starved block gains +6, +11 and +16 windows of history changes the velocity-subset mAP of S5-B by 3.73 ± 0.35 , 4.91 ± 0.50 and 5.23 ± 0.49 points, and that of S5-S by 3.54 ± 0.39 , 4.29 ± 0.52 and 4.81 ± 0.52 . The zero-shift control, which re-runs the release’s own boundary placement through the same pipeline, returns 0.00. The three rungs share their frames, so a frame-difficulty confound aligned with chunk position cannot produce the gradient. Confidence on detections matched to a label rises by 0.0409 and 0.0519, and total output falls by 3.49 and 5.09 boxes per frame, so both terms of precision move: the starved arm emits more boxes and scores the correct ones lower. Tab. 2 lists all four amounts.

The observational contrast across chunk positions. The released evaluation already contains the contrast, without any intervention. Reconstructing chunk position over the 20 296 labeled frames of the released evaluation and scoring every labeled box, S5-B reaches 34.445 mAP on the frames given 1–4 windows and 40.226 on those given 17–21 (Fig. 1, top; Tab. 1). This contrast compares different frames, so it is differenced against the three RVT checkpoints, evaluated on the same frames with a state that crosses chunk boundaries. That controls for frame-position effects shared with them. It leaves 5.32 ± 0.53 points for S5-B ($z = 10.0$) and 4.37 ± 0.61 for S5-S ($z = 7.2$), against placebo differences among the RVT checkpoints of at most 0.67 points ($z = 1.56$). A permutation null reshuffles chunk position among the frames of each sequence, which preserves the scene, the sequence lengths, the label density and the block sizes and destroys only the correspondence between a frame and its history. Over 200 reshuffles the null has a standard deviation of about 0.55 points; both SSM values lie outside every reshuffle ($p = 0.005$, the smallest attainable), and every placebo stays within 1.52 standard deviations. A label-free statistic shows the same split: the mean confidence of S5-S rises from 0.463 at 1 window to

	1–4 win.	17–21 win.	raw Δ	net of RVT
RVT-t	38.180	38.194	0.01	-0.67 ± 0.43
RVT-s	38.371	39.167	0.80	0.51 ± 0.48
RVT-b	41.561	42.126	0.56	0.16 ± 0.55
S5-S	31.578	36.410	4.83	4.37 ± 0.61
S5-B	34.445	40.226	5.78	5.32 ± 0.53

Table 1. Released evaluation, no intervention: mAP over every labeled box of the 20 296 frames, by the recurrent history the release gave the frame. Raw Δ is the second column minus the first. The last column subtracts the RVT trend (the mean of the other two RVT rows for an RVT row, of all three for an S5 row), $\pm$ one cluster-bootstrap standard error over 406 sequences. The RVT rows are placebos: their state crosses chunk boundaries.

0.574 at 17, while that of RVT-t stays flat. The observational and interventional readings agree in sign and magnitude, and the paired intervention carries the claim because it holds the frames fixed (Supplement Sec. 1).

RVT under the same intervention. That differencing treats the RVT checkpoints as a control because their state survives a chunk boundary. The same-frame intervention applied to RVT measures that property. Driven on the SSM chunk grid, RVT reproduces its stored per-window detections bit for bit at zero shift, and the same 3 574 frames are re-scored with every boundary moved 5 windows later. With the released state carried across the boundary the paired change is -0.01 ± 0.03 points for RVT-s ($z = -0.3$) and 0.05 ± 0.04 for RVT-b ($z = 1.3$). Dropping the state at every chunk start, the condition the SSM release is already in, moves the same weights on the same frames by 5.39 ± 0.45 ($z = 12.0$) and 6.91 ± 0.64 points ($z = 10.9$). Random-frame nulls are centered within 0.112 points of zero. The carried arm shows that the control the difference-in-differences relies on holds as a measurement. The reset arm shows that the shift reaches the model, so the null of the carried arm is a property of the released evaluation and not of an intervention that failed. One model, one set of weights and one set of frames give a zero or a 5.39-point effect depending only on whether the state survives the boundary, a variable that no protocol record contains (Supplement Sec. 1).

Controls. Each alternative reading of the effect has its own null. A pipeline artifact is excluded by the zero-shift control (0.00) and by the bit-identical reproduction of the stored RVT detections. Frame difficulty is excluded by pairing within frame, by the dose response on one frame set, and by the permutation null. An evaluator artifact is excluded by the label-free confidence shift and by rescoring every detection dump with `pycocotools`, which agrees with our evaluator to within 0.101 points on all five checkpoints. An architecture-specific reading is excluded by the

<i>SSM-ViT, history gained by the starved block (same 3 574 frames)</i>		
	S5-B	S5-S
+0 (control)	0.00	0.00
+6 windows	3.73 ± 0.35	3.54 ± 0.39
+11 windows	4.91 ± 0.50	4.29 ± 0.52
+16 windows	5.23 ± 0.49	4.81 ± 0.52
<i>RVT, boundary moved 5 windows (same 3 574 frames)</i>		
state at boundary	RVT-s	RVT-b
carried (released)	-0.01 ± 0.03	0.05 ± 0.04
dropped	5.39 ± 0.45	6.91 ± 0.64

Table 2. Same-frame intervention on the velocity-evaluable subset: Δ mAP (points, long-history arm minus short-history arm, $\pm$ one cluster-bootstrap standard error over 406 sequences). Top: the SSM-ViT checkpoints at four boundary amounts, the last row being the intervention of Fig. 1. Bottom: RVT under the same boundary shift with its state carried and with its state dropped at every chunk start.

two RVT state regimes. Dependence between frames of a sequence is handled by resampling sequences, not frames, in every interval.

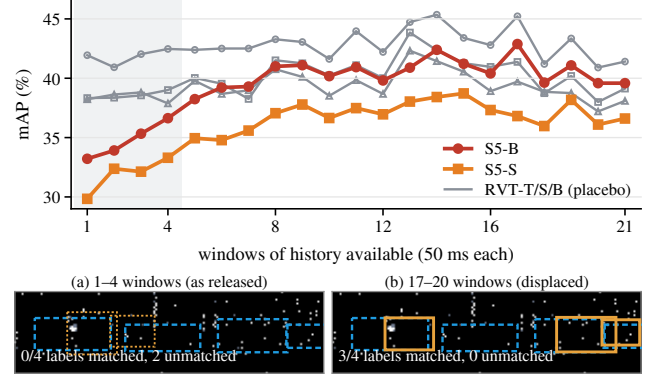
What the pooled score reports. One released number pools detections whose recurrent history differs by a factor of 21. Restricting evaluation to the frames that receive 17–21 windows gives an S5-B score 1.28 points above the pooled released score, and 0.86 points for S5-S. Two systems compared under this protocol at the same nominal timestamps need not have been given the same temporal evidence, and the protocol record contains no variable by which a reader could tell.

5. What else the timestamp leaves implicit

Recurrent history is one component of the support behind a score. Two audits measure the others on released artifacts, the input timing inside the newest window and the acquisition timing of the label. Each is a timing descriptor that the nominal timestamp does not encode. Their derivations, figures and tables are in Supplement Secs. 5 and 14.

Protocol record. Call a quantity *protocol-identifiable* when the protocol record of Sec. 1 determines it uniquely. The input window and the published exposure interval are in the record. The effective predictor timing, the recurrent history pooled in a score, and the attribution of event timing within an exposure are not.

Predictor side (Gen1). The released RVT configuration [21] indexes each ten-bin, 50 ms input window by its end, and places that end on a label time. Ablating each bin of the released RVT-t checkpoint over 480 Gen1 validation samples puts the ablation-sensitivity centroid at -23.81 ms, so the evidence of the newest window is centered about half a window before the instant the label asserts. A regression



of detection-center residuals on label velocity, which carries the temporal calibration of [6] to a box center and is fitted on nominal-time-matched detections, gives a coefficient of -2.40 ± 7.18 ms, so the state the output reports is the one at the label time. One timestamp thus stands for two instants whose point estimates are 26.2 ms apart, the centroid of the evidence and the time of the state the output reports. All five released checkpoints show the same newest-window centroid to within a fraction of one bin. Over a 1000 ms horizon the newest window accounts for only 22.7 % of the summed window-ablation sensitivity. Most of the evidence behind a detection lies in the recurrent history, which is the component Sec. 4 manipulates (Supplement Secs. 5, 6 and 8).

Label side (DSEC). Across the eighteen DSEC training sequences whose exposure files resolve [13] (21 142 frames), published exposure width varies by a factor of 127 against a frame period fixed at 50 ms, and 32.5 % of frames sit at the auto-exposure ceiling. A label on the frame clock therefore summarizes acquisition intervals that differ by two orders of magnitude. Inside one ceiling exposure the event rate moves by a factor of 3.0 between its extremes, against 1.1 in a daytime control, and its spectrum carries the 100 Hz line of rectified mains lighting. Across all six ceiling sequences and 16 516 labeled boxes, 95.4–100.0 % of boxes

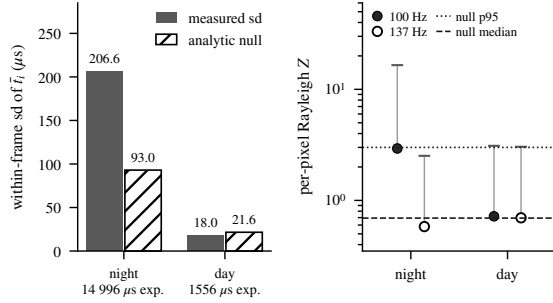


Figure 2. Label-side audit on DSEC. Left: within-frame dispersion of the per-box event-time centroid inside one published exposure against its analytic uniform null, for 1134 frames of a ceiling-exposure sequence (14 996 μ s) and 20 frames of a daytime control (1556 μ s). Right: per-pixel Rayleigh statistic at 100 Hz and at the 137 Hz off-frequency control for the same two sequences. Details in Supplement Secs. 14–16.

per sequence carry that signature above an off-frequency control on the same events. Within-box dispersion of event time inside one exposure is 206.6 μ s against an analytic null of 93.0 μ s (Fig. 2). Within-exposure event timing contains an illumination-locked component and cannot be attributed to object timing without separating that component first (Supplement Secs. 14 and 16). DSEC-Det ships more than one annotation version [15], and Supplement Sec. 15 names the one measured here.

6. Support-Conditioned AP

Sec. 4 shows that one released protocol pools detections that received different histories. Criterion 2 of Sec. 3 says what to fix. This section scores the five released Gen1 checkpoints at a declared history and compares the result with the pooled score.

Protocol. For a labeled frame scored at window r and a history length H , the detector starts from a reset recurrent state at window $r - H + 1$ and runs through window r , and the detections of window r are scored with the labels, confidence threshold, NMS and evaluator of the released-protocol dumps. We call the resulting AP at fixed H Support-Conditioned AP. H counts the windows read up to and including the scored window; we write H_{nominal} for this requested length. A frame with fewer than H windows available receives $H_{\text{forced}} = \min(H, r + 1)$ windows instead. All checkpoints are scored on the same 3 000 frames. They are a stratified sample of the 18 947 labeled frames that have at least 80 windows available, so that $H_{\text{forced}} = H_{\text{nominal}}$ at every $H \in \{1, 5, 10, 21, 40, 80\}$. The eligible frames lie in 405 of the 406 validation sequences. The sequence is the stratum: allocation is proportional to a sequence’s eligible frames, every such sequence contributes at least one frame and at most 36, and the draw used a

fixed seed and was made before the runs at $H = 40$ and 80. The column *pooled* is the released evaluation of each checkpoint, in which the history depends on the release and on the frame, restricted to the same 3 000 frames. The access log of every run shows that the number of distinct windows read equals H_{forced} and that the first window read is $r - H_{\text{forced}} + 1$ in every cell of every checkpoint (Supplement Sec. 1.4). The protocol reuses released weights and labels and adds no training.

Result. Table 3 lists the scores. On the 3 000 frames the pooled ordering is RVT-b, RVT-s, RVT-t, S5-B, S5-S. At $H = 1$ the ordering is S5-B, S5-S, RVT-b, RVT-s, RVT-t, at $H = 21$ it is S5-B, RVT-b, S5-S, RVT-s, RVT-t, at $H = 40$ it is RVT-b, S5-B, RVT-s, RVT-t, S5-S, and at $H = 80$ it equals the pooled ordering. Kendall’s τ between the pooled ordering and the orderings at $H = 1, 5, 10, 21, 40$ and 80 is $-0.20, 0.00, -0.20, 0.00, 0.60$ and 1.00. S5-B is first at each $H \in \{1, 5, 10, 21\}$. Its gap to every other checkpoint has a 95 % interval that excludes zero at $H = 1, 5$ and 10, and at $H = 21$ for every checkpoint except RVT-b (Tab. 4 and Supplement Sec. 1.4). The pooled ordering is reproduced in 46.5 % of draws on these frames and in 42.5 % over all labeled boxes (Supplement Sec. 4), while the full orderings at $H = 1, 5, 10, 21, 40$ and 80 are reproduced in 67.0, 70.5, 52.0, 54.0, 60.0 and 66.5 % of draws.

Several pairwise gaps change sign. RVT-t minus S5-S is $+3.34 [+1.94, +4.56]$ points in the pooled score and $-2.88 [-4.65, -1.78]$ at $H = 1$. RVT-s minus S5-S is $+3.72 [+2.25, +4.93]$ pooled and $-2.42 [-4.18, -1.18]$ at $H = 1$. RVT-b minus S5-B is $+3.13 [+1.82, +4.39]$ pooled and $-4.15 [-6.07, -2.80]$ at $H = 1$, stays negative at $H = 5, 10$ and 21 ($-1.33 [-2.61, +0.06]$ at $H = 21$), and is positive at $H = 40 (+1.28 [-0.11, +2.53])$ and $H = 80 (+4.79 [+3.52, +5.85])$. The ranking of these two checkpoints therefore crosses between $H = 21$ and $H = 40$, at $H \approx 31$ if the gap is interpolated linearly between the two scored values. The interval of the gap excludes zero at $H = 1, 5, 10$ and 80 and includes it at $H = 21$ and $H = 40$. At $H = 80$ each RVT checkpoint is above each S5 checkpoint with an interval that excludes zero, whereas at every $H \leq 21$ S5-B is above every RVT checkpoint. For RVT-s and RVT-t against S5-B the pooled intervals include zero ($+0.57 [-1.13, +2.02]$ and $+0.19 [-1.43, +1.91]$), so the pooled score cannot separate them, while at every $H \leq 21$ the point gaps lie between -6.06 and -3.43 and between -6.53 and -3.83 and the intervals exclude zero. Fixing the support thus resolves comparisons that the pooled score leaves open, and which checkpoint is ahead depends on the support at which it is scored. Tab. 4 lists the gaps for all pairs in the pooled score and at $H = 1, 21, 40$ and 80 (Supplement Sec. 1.4 lists all seven columns)

	pooled	$H = 1$	$H = 5$	$H = 10$	$H = 21$	$H = 40$	$H = 80$
RVT-t	39.77 [37.62, 42.66]	28.58 [26.33, 31.21]	34.15 [31.88, 37.01]	36.09 [33.67, 38.84]	37.04 [34.86, 39.72]	38.19 [35.87, 40.82]	38.80 [36.56, 41.72]
RVT-s	40.15 [38.38, 42.70]	29.04 [26.99, 31.91]	34.60 [32.47, 37.36]	35.94 [33.95, 38.68]	37.44 [35.34, 40.36]	38.44 [36.30, 41.24]	38.93 [37.00, 41.62]
RVT-b	42.71 [41.05, 44.91]	30.96 [28.86, 33.52]	36.45 [34.57, 38.94]	38.17 [36.33, 40.56]	39.53 [37.68, 41.92]	40.57 [38.71, 42.96]	41.22 [39.31, 43.58]
S5-S	36.43 [34.44, 39.26]	31.45 [29.44, 34.31]	35.58 [33.76, 38.52]	36.78 [34.87, 39.70]	37.89 [35.97, 40.78]	35.69 [33.74, 38.87]	30.91 [28.87, 33.89]
S5-B	39.58 [37.86, 41.74]	35.10 [33.17, 37.63]	38.87 [37.36, 41.09]	40.20 [38.55, 42.42]	40.87 [39.01, 43.06]	39.28 [37.78, 41.83]	36.43 [34.91, 38.90]
Kendall τ against pooled	1.00	-0.20	0.00	-0.20	0.00	0.60	1.00

Table 3. mAP (points) of the five released Gen1 checkpoints on the same 3 000 frames, over every labeled box, with the recurrent history fixed to H windows (reset state, exactly H windows read) or as given by the released evaluation (pooled). Brackets: 95 % interval from 200 cluster-bootstrap draws over the 405 sequences, the same draw for every checkpoint and column. The final row gives Kendall’s τ between the ordering of each column and the pooled ordering.

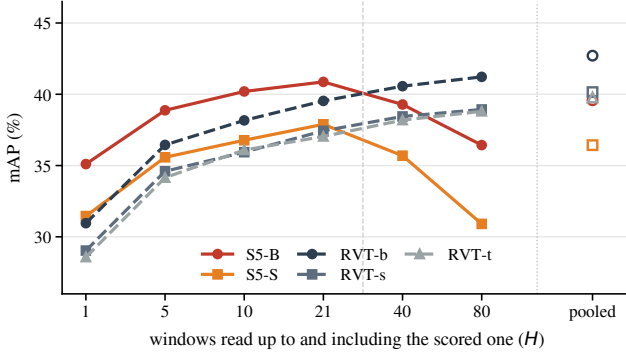


Figure 3. mAP of the five checkpoints on the same 3 000 frames at $H = 1, 5, 10, 21, 40, 80$ (filled markers, equal spacing on the axis) and in the pooled released evaluation (open markers). The dashed line marks the chunk length of the released SSM-ViT evaluation. RVT-t and S5-B differ by +0.19 points at the pooled position. Intervals are in Tab. 3 and the paired gaps in Tab. 4.

and Fig. 3 plots the curves over H .

What Support-Conditioned AP equalizes. The access log confirms that exactly H windows are read. Replacing the oldest admitted window by zeros changes the output in 2251 to 2254 of 2259 probed cells per checkpoint at $H = 21, 40$ and 80 , and in every other probed cell the window already holds no events or the frame has no detection (Supplement Sec. 1.4). The full admitted history therefore reaches the detection, and the support is the interval of Criterion 2. Every checkpoint starts from the same reset state. The protocol therefore equalizes the two things the pooled score leaves free, the reset policy and the history length. Under the released evaluations these differ by family: the RVT release carries its state across a sequence, while the SSM-ViT release scores each chunk from a reset state (Sec. 4). This is why the pooled RVT scores sit above their $H = 21$ scores (RVT-b: 42.71 against 39.53) and the pooled S5 scores sit below theirs (S5-B: 39.58 against 40.87). The pooled column compares checkpoints that were given different support. Table 3 compares them at the same support, which is the comparison a timing benchmark is taken to make. The unit compared is the released checkpoint under one evaluation, the unit a leaderboard ranks.

The curve over H as a result. Support-Conditioned AP reports a curve where the pooled protocol reports one number, and the curve carries information the number hides (Fig. 3). From $H = 1$ to $H = 21$ the RVT checkpoints gain between 8.4 and 8.6 points and the S5 checkpoints between 5.8 and 6.4, so at $H = 1$ both S5 checkpoints rank above all three RVT checkpoints and the families converge as history grows. Beyond $H = 21$ the RVT scores keep rising, by a further 1.5 to 1.8 points up to $H = 80$, while the S5 scores fall, by 4.4 points for S5-B and 7.0 for S5-S. The S5-S score at $H = 80$ (30.91) is below its score at $H = 1$ (31.45). The S5 scores at $H = 40$ and 80 come from chunks longer than the 21 windows of the released evaluation, and the Gen1 training configurations of both repositories set the sequence length to 21. We did not test whether this accounts for the decline. A reader choosing a detector for a short-latency budget and a reader choosing one for long sequences would pick differently from the same table, S5-B for $H \leq 21$ and RVT-b for $H \geq 40$, and neither choice can be read from the pooled column.

Reporting recipe. The protocol reduces to five steps that apply to any recurrent detector with a released checkpoint. (i) Declare a set of history lengths; we use $H \in \{1, 5, 10, 21, 40, 80\}$, where 21 is the chunk length of the released streaming evaluation and 40 and 80 exceed it. (ii) Restrict scoring to the frames that have the largest H available, sample them if the largest H is costly, and state the frame count and the sampling. (iii) For every frame and every H , run from a reset state over exactly H windows and keep the detections of the last. (iv) Log the windows the loader returned and probe the oldest one, so that the declared support is the realized one. (v) Report the fixed- H scores with paired sequence-bootstrap intervals beside the pooled score, together with the distribution of history lengths the pooled score contains.

What a benchmark should record. A nominal timestamp does not determine the effective temporal support or the label-acquisition support, and the intervention above shows a case in which the unrecorded support changes the score. We propose that a benchmark record, for each scored

pair (first – second)	pooled	$H = 1$	$H = 21$	$H = 40$	$H = 80$
RVT-t – RVT-s	-0.38 [-1.22, +0.45]	-0.46 [-1.49, +0.35]	-0.40 [-1.63, +0.37]	-0.26 [-1.28, +0.47]	-0.13 [-1.11, +0.74]
RVT-t – RVT-b	-2.93 [-3.87, -2.02]	-2.38 [-3.20, -1.72]	-2.49 [-3.77, -1.56]	-2.38 [-3.42, -1.45]	-2.42 [-3.59, -1.41]
RVT-t – S5-S	+3.34 [+1.94, +4.56]	-2.88 [-4.65, -1.78]	-0.85 [-2.58, +0.22]	+2.49 [+0.77, +3.64]	+7.89 [+6.14, +9.31]
RVT-t – S5-B	+0.19 [-1.43, +1.91]	-6.53 [-8.82, -4.91]	-3.83 [-5.44, -2.20]	-1.10 [-2.85, +0.67]	+2.37 [+0.83, +3.66]
RVT-s – RVT-b	-2.56 [-3.59, -1.60]	-1.92 [-2.54, -1.03]	-2.10 [-2.93, -1.14]	-2.12 [-2.95, -1.30]	-2.29 [-3.17, -1.40]
RVT-s – S5-S	+3.72 [+2.25, +4.93]	-2.42 [-4.18, -1.18]	-0.46 [-1.86, +0.96]	+2.75 [+1.21, +4.05]	+8.03 [+6.45, +9.53]
RVT-s – S5-B	+0.57 [-1.13, +2.02]	-6.06 [-8.07, -4.43]	-3.43 [-4.91, -1.75]	-0.84 [-2.19, +0.62]	+2.50 [+1.15, +3.71]
RVT-b – S5-S	+6.28 [+5.04, +7.22]	-0.50 [-2.27, +0.43]	+1.64 [+0.55, +2.79]	+4.87 [+3.73, +5.92]	+10.31 [+8.84, +11.58]
RVT-b – S5-B	+3.13 [+1.82, +4.39]	-4.15 [-6.07, -2.80]	-1.33 [-2.61, +0.06]	+1.28 [-0.11, +2.53]	+4.79 [+3.52, +5.85]
S5-S – S5-B	-3.15 [-4.02, -1.99]	-3.65 [-4.76, -2.41]	-2.98 [-4.01, -1.93]	-3.59 [-5.01, -2.25]	-5.52 [-6.88, -4.15]

Table 4. Paired mAP gaps (points) between the five released Gen1 checkpoints on the same 3 000 frames, with 95 % intervals from 200 bootstrap draws over the 405 sequences. Pooled is the released evaluation restricted to the same frames. Gaps are computed from unrounded scores. A gap whose interval excludes zero is a difference the sequence resampling does not remove. The sign of the pooled gap and the sign at $H = 1$ are opposite, with both intervals excluding zero, for RVT-t – S5-S, RVT-s – S5-S and RVT-b – S5-B.

field	RVT release	SSM-ViT release	Support-Cond. AP
nominal times-tamp	recorded	recorded	recorded
input window	recorded	recorded	recorded
state-reset policy	sequence start only; not recorded	every 21-window chunk; not recorded	at window $r - H + 1$; declared
effective history length	up to sequence length; not recorded	1–21 windows; not recorded	H windows; logged and probed

Table 5. Four of the five fields of Sec. 6 for the two released Gen1 evaluations and for Support-Conditioned AP. A field marked not recorded is not part of what the protocol reports with a score. The fifth field, label-acquisition support, belongs to the dataset and is audited on DSEC in Sec. 5.

sample, five fields. The *nominal timestamp* is the instant at which the sample is scored. The *input window* is what the detector reads at that timestamp. The *state-reset policy* states where the recurrent state is cleared, including chunk boundaries and sequence starts. The *effective history length* is the number of windows read, as the access log of the evaluation run records it. The *label-acquisition support* is the interval over which the label’s evidence was acquired. By Criterion 2, the reset policy and the history length determine the effective temporal support. Tab. 5 fills in these fields for the two releases examined here. A score is then reported at a declared history length, as in Tab. 3. The record makes support checkable, and Support-Conditioned AP makes it equal, so that scores are compared at a history that both detectors received.

7. Conclusion

A benchmark timestamp does not determine the temporal evidence behind the score reported under it. With frames, weights, labels, evaluator and evaluated timestamps fixed, moving only the chunk boundary shifts S5-B by 4.20 mAP over all labeled validation boxes and by 5.23 mAP on the velocity-evaluable subset. The chunk boundary changes the recurrent support behind a reported score while every recorded quantity stays fixed, which is the witness pair that refutes temporal-support identifiability. RVT shows the

same effect as soon as its state is reset at the boundary and none while it is carried, so the effect belongs to the reset policy and not to one architecture.

Support-Conditioned AP removes this degree of freedom by fixing the reset state and the history behind every score. On five released Gen1 checkpoints, S5-B moves from fourth in the pooled score to first at every $H \in \{1, 5, 10, 21\}$, RVT-b overtakes it between $H = 21$ and $H = 40$, Kendall’s τ between the pooled and fixed- H orderings is at most 0.00 for $H \leq 21$, and three pairwise gaps reverse sign between the pooled score and $H = 1$ with intervals that exclude zero on both sides.

Two scores at the same nominal timestamp are comparable in time only when the support behind them is the same. A benchmark can make this checkable by recording five fields for each scored sample (nominal timestamp, input window, state-reset policy, effective history length, label-acquisition support), and a paper can make it hold by reporting scores at a declared history length.

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